The Dawkins-Hayek Convergence: Spontaneous Order as Extended Phenotype

Two Independent Solutions to the Problem of Order Without Design

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Abstract

This paper argues that Charles Darwin, Richard Dawkins, and the Austrian economists Carl Menger, Ludwig von Mises, and F.A. Hayek independently solved the same fundamental problem: explaining complex functional order without invoking a designer.
The Darwinian solution rejected teleology and group-level explanations in favor of gene-level selection and extended phenotypes.
The Austrian solution rejected constructivist rationalism and state design in favor of spontaneous order emerging from individual action. I argue that these are not mere analogies but structurally identical solutions to formally equivalent problems.
The convergence reveals a deep theoretical unity with profound implications for institutional analysis. Specifically, Dawkins's concept of the extended phenotype completes Hayek's concept of extended order by providing the missing mechanistic foundation: legal institutions are extended phenotypes of competing legal memes.
This synthesis undermines both multilevel selection theory in biology and constructivism in social science, as both reintroduce the teleological errors their respective traditions sought to eliminate. The paper concludes by outlining implications for constitutional political economy and institutional design.

Keywords: spontaneous order, extended phenotype, gene-level selection, Austrian economics, memetics, institutional evolution

JEL Classifications: B25, B53, D02, K00, P48

I. Introduction: The Problem of Order Without Design

Complex functional order surrounds us. Organisms exhibit intricate adaptations: the eye focuses light, the heart pumps blood, the kidney filters toxins. Social institutions exhibit equally intricate coordination: markets allocate resources, legal systems resolve disputes, languages enable communication. The temptation to explain such order by

invoking a designer is overwhelming. Before Darwin, naturalists explained biological adaptation through divine design or teleological principles inherent in nature itself. Before Carl Menger, economists explained institutional order through state design or the conscious planning of enlightened rulers.

Both explanations failed. The biological version collapsed under Darwinian natural selection. The economic version collapsed under the Austrian critique of constructivist rationalism. What is less widely recognized is that these collapses represent solutions to the same problem using the same conceptual architecture. The functional order of organisms and the functional order of institutions both required explanations that eschewed designers, purposes, and teleology while still accounting for the appearance of design.

Charles Darwin solved the biological problem by showing how differential survival and reproduction could generate apparent design without a designer (Darwin 1859). His insight was radicalized by George Williams and Richard Dawkins, who demonstrated that the relevant unit of selection is neither the organism nor the group but the gene (Williams 1966; Dawkins 1976). Dawkins extended this insight further with the concept of the extended phenotype: genes produce effects beyond the organism's body, and these extended effects are themselves subject to selection (Dawkins 1982). The beaver's dam, the spider's web, the caddis fly larva's stone house all constitute extended phenotypes of the genes that build them.

Carl Menger solved the economic problem by showing how complex institutions like money could emerge without anyone designing them (Menger 1871). His insight was developed by Ludwig von Mises and F.A. Hayek into a general theory of spontaneous order (Mises 1949; Hayek 1973, 1988). Hayek distinguished between taxis, or designed order imposed from above, and kosmos, or spontaneous order emerging from individual actions. He coined the term "extended order" to describe the civilization-wide coordination that arises from individuals following abstract rules without understanding the overall pattern they generate. Hayek explicitly noted that this extended order resulted "not from human design or intention but from spontaneously conforming to certain traditional and largely moral practices" (Hayek 1988, p. 6).

The terminological parallel between "extended phenotype" and "extended order" is not coincidental. Both concepts address the same theoretical challenge: explaining functional order at a level of organization higher than the acting unit without invoking group-level purposes or design. Dawkins explains how genes produce phenotypic effects beyond the organism's boundaries. Hayek explains how individual actions produce institutional effects beyond the actor's intentions. Both reject teleological explanations at the higher level. Both locate the generative mechanism at the lower level.

This paper argues that the Dawkins-Hayek convergence is not metaphorical but structural. The two theories share the same logical architecture because they solve formally equivalent problems. Section II reconstructs the Darwinian solution, emphasizing Williams's and Dawkins's gene-centered view and Dennett's criterion of cui bono. Section III reconstructs the Austrian solution, emphasizing Menger's account

of institutional emergence and Hayek's extended order. Section IV demonstrates the structural identity between the two solutions through systematic comparison. Section V synthesizes the two traditions by showing how Dawkins's concept of the extended phenotype completes Hayek's concept of extended order: institutions are extended phenotypes of memes competing for replication. Section VI explores implications for multilevel selection theory, constructivism, and institutional design.

The stakes are high. If the argument succeeds, it undermines both group selection in biology and constructivism in social science as resurrections of pre-Darwinian and pre-Mengerian errors. It also provides a mechanistic foundation for analyzing legal evolution, constitutional change, and institutional competition without appealing to group-level functions or purposes. The synthesis developed here establishes the theoretical groundwork for what I call Extended Phenotype Theory (EPT), a research program applying gene-centered evolutionary logic to legal systems.

II. The Darwinian Solution

The Problem: Apparent Design Without a Designer

Pre-Darwinian biology faced an explanatory crisis. Organisms exhibit exquisite functional organization. The vertebrate eye possesses a lens that focuses light, a retina that transduces photons into neural signals, and an iris that regulates light intensity. The heart consists of chambers, valves, and muscular walls precisely configured to pump blood. The kidney filters blood through glomeruli, reabsorbs useful solutes in tubules, and excretes waste. Every species manifests countless such adaptations. The temptation to explain this functional complexity through conscious design or inherent teleology proved nearly irresistible.

William Paley's watchmaker argument epitomized the design inference (Paley 1802). Finding a watch on the heath, one infers a watchmaker. Finding an eye, one should infer a designer. The eye's functional organization demands explanation, and design provides it. Absent Darwin, this argument was compelling. Even Kant, who rejected theological proofs of God's existence, retained teleological principles in biology, distinguishing organisms from mere mechanisms by their self-organizing purposiveness (Kant 1790).

Darwin's solution was conceptually simple but psychologically difficult. Heritable variation exists in every population. Differential survival and reproduction select variants that enhance survival and reproduction. Across generations, this process accumulates incremental changes. Complex adaptations emerge through innumerable small steps, each individually advantageous (Darwin 1859). The appearance of design arises from cumulative selection operating over vast time scales. No designer is necessary. No teleological principle guides the process. Adaptation is the unintended consequence of differential gene survival.

Williams's Critique: Adaptation as Gene-Level Phenomenon

Despite Darwin's breakthrough, biologists continued invoking group-level and species-level explanations. Animals were said to evolve traits "for the good of the species." Population control mechanisms supposedly prevented overexploitation of resources. V.C. Wynne-Edwards argued that animals evolved altruistic restraint to preserve group welfare (Wynne-Edwards 1962). George Williams demolished this edifice (Williams 1966).

Williams argued that adaptation must be explained at the level where selection operates most forcefully. Group selection is weak because groups reproduce slowly and migrate frequently. Individual selection is stronger because individuals reproduce faster. But gene selection is strongest because genes can outlast both individuals and groups. A gene that enhances its own replication will spread even if it harms the organism or the group. Selection at the gene level trumps selection at higher levels.

Williams introduced a key methodological principle: invoke adaptation only when necessary and locate it at the lowest possible level. Apparent group-level adaptations usually dissolve into individual-level or gene-level phenomena upon closer inspection. Population control, for instance, results from individual organisms maximizing their own fitness subject to ecological constraints, not from self-sacrificial restraint for group benefit (Williams 1966, pp. 93-96).

This gene-centered view revolutionized evolutionary biology. It explained puzzling phenomena like senescence (genes that enhance early reproduction spread even if they cause late-life decline), sexual conflict (male and female reproductive interests diverge), and genomic conflicts (genes at different loci compete within the same organism). The gene's-eye view revealed that organisms are not unified agents pursuing their own survival but coalitions of genes pursuing replication. Adaptation serves gene replication, not organism survival or species preservation.

Dawkins's Radicalization: The Extended Phenotype

Richard Dawkins radicalized Williams's insight. In *The Selfish Gene*, Dawkins argued that genes are replicators and organisms are vehicles (Dawkins 1976). Genes replicate with high fidelity across generations. Organisms are temporary constructions through which genes propagate. Natural selection operates on replicators, not vehicles. Organisms exist because genes build them as survival machines. The organism-centered view is psychologically natural but theoretically backwards. The correct starting point is the gene.

Dawkins extended this logic beyond the organism's body in *The Extended Phenotype* (Dawkins 1982). A phenotype is any effect of a gene. Typically, phenotypes are traits of the organism carrying the gene: eye color, wing shape, metabolic rate. But genes also produce effects outside the organism's body. The beaver builds a dam. The spider weaves a web. The caddis fly larva constructs a protective case from stones and twigs. These structures are extended phenotypes: phenotypic effects of genes expressed in environmental modifications rather than bodily traits.

The extended phenotype concept has radical implications. It severs the connection between genes and organism boundaries. A gene for dam-building affects the beaver's environment, not merely the beaver's body. Selection operates on the gene based on the dam's effects on gene replication. The dam is a phenotype of beaver genes just as surely as the beaver's teeth and tail. The organism ceases to be the privileged locus of phenotypic expression. Genes reach beyond organism boundaries to manipulate the world.

The concept applies to interspecific interactions. A parasite manipulates its host's behavior. The liver fluke (*Dicrocoelium dendriticum*) infects ants as an intermediate host. Infected ants climb grass blades at night, increasing the probability of ingestion by grazing sheep, the fluke's final host. The ant's behavior is a phenotype of fluke genes, not ant genes (Dawkins 1982, pp. 163-164). Asking what benefit the ant derives from climbing grass is the wrong question. The ant derives no benefit. The behavior benefits fluke genes by facilitating transmission to the final host.

Daniel Dennett crystallized the methodological implication: adaptation requires identifying the beneficiary (Dennett 1995). *Cui bono?* Who benefits? A benefit by itself is not explanatory. Benefits abound in nature. Flowers benefit bees with nectar. Trees benefit squirrels with acorns. But these benefits do not explain floral or arboreal traits unless they enhance the replication of genes in flowers or trees. The nectar benefits the flower by attracting pollinators who transfer pollen. The acorn benefits the tree by inducing squirrels to bury seeds that germinate. Benefits to bees and squirrels are byproducts. The explanation runs through plant genes, not pollinator genes.

Dennett writes: "*A benefit by itself is not explanatory; until it can be shown how the benefit actually redounds to enhance the replicative power of a replicator, it just sits there, incapable of explaining anything*" (Dennett 1995, p. 477). This principle disciplines adaptationist reasoning. Every purported adaptation must be connected to the replication of specific genes. Group benefits do not explain adaptations unless they enhance individual or gene-level replication. The appearance of teleology at higher levels dissolves into mechanistic gene competition at lower levels.

Application to Institutions: Dysfunctional Norms as Extended Phenotypes

The cui bono test reveals that many social institutions persist despite generating manifest harm to societies. Jim Crow laws in the United States (1877-1965) provide a stark example. These laws enforced racial segregation in the South, systematically denying African Americans political rights, economic opportunity, and social equality. The laws impoverished the region, retarded economic development, and inflicted massive human suffering. Yet they persisted for nearly a century. A group-level functional explanation fails: the laws did not benefit Southern society as a whole. The explanation lies at the meme level.

White supremacist memes benefited from Jim Crow institutions. These memes replicated through families, churches, schools, and political organizations. The institutional structure was the extended phenotype of these memes: segregated schools taught racial hierarchy, courts enforced racial codes, legislatures enacted racial

restrictions. Each component reinforced meme transmission. Individuals who adopted and promoted white supremacist memes gained social status, political power, and economic advantage within the segregated system. The memes replicated because they benefited their carriers, not because they benefited Southern society.

The termination of Jim Crow illustrates memetic competition. Civil rights memes (equal dignity, constitutional rights, color-blind law) competed with white supremacist memes. These rival memes spread through courts (Brown v. Board of Education), legislation (Civil Rights Act of 1964), and social movements (Freedom Riders, Montgomery bus boycott). The federal government disrupted the institutional extended phenotype: desegregating schools, enforcing voting rights, prohibiting discrimination. As the institutional structure collapsed, white supremacist memes lost their replication advantage. Civil rights memes displaced them not because society benefited but because the memetic environment shifted. The cui bono test points to competing memes and their institutional phenotypes, not group-level functions.

Argentine constitutional law provides a parallel case. Article 14bis, inserted in 1957, mandates comprehensive labor rights: "just remuneration," "participation in company profits," "protection against arbitrary dismissal," and "extensive social security." The article entrenched a corporatist labor regime that generated chronic unemployment, inflated labor costs, and incentivized informality. By 2024, over 40% of Argentine workers operate outside formal employment to evade Article 14bis mandates. The regime has manifestly harmed Argentine society through reduced employment, lower productivity, and institutional instability. Yet Article 14bis persists, insulated from reform by constitutional rigidity and political entrenchment.

A group-level explanation fails. The article does not benefit Argentine society. The explanation is memetic. Corporatist labor-law memes benefit from Article 14bis. These memes replicate through law schools, labor tribunals, union bureaucracies, and political parties. Each institution is an extended phenotype of the corporatist meme complex. Judges trained in labor law apply protective doctrines. Law professors teach students that labor rights are fundamental. Union leaders defend the regime. Politicians campaign on preserving Article 14bis. Individuals who adopt and transmit these memes gain professional credentials, judicial appointments, union positions, and political support. The memes replicate because they benefit their carriers within the institutional structure, not because they benefit Argentina.

These examples vindicate Dennett's cui bono criterion applied to institutional analysis. Dysfunctional institutions persist because they are extended phenotypes of memes that benefit from their existence. Asking whether society benefits is the wrong question. The correct question is: which memes benefit, and through what mechanisms do those memes replicate? Jim Crow benefited white supremacist memes. Article 14bis benefits corporatist labor-law memes. Both institutions persisted despite societal harm because memetic replication operates independently of group welfare. This analysis demolishes any claim that institutions exist to serve collective functions. They exist because memes build them and replicate through them.

The Rejection of Group Teleology

The gene-centered view undermines group selection and multilevel selection theory (MLS2). Group selectionists argue that groups compete, and traits benefiting groups can evolve even if they harm individuals (Sober and Wilson 1998; Wilson and Wilson 2007). A group of altruists outcompetes a group of selfish individuals. Altruism spreads because altruistic groups proliferate. This resurrection of group-level teleology contradicts Williams's and Dawkins's core insight.

Dawkins's response is unequivocal. Groups do not replicate with sufficient fidelity to function as units of selection. Individuals migrate between groups. Groups fission and fuse. Group boundaries are porous and arbitrary. In contrast, genes replicate with high fidelity, survive across generations, and serve as coherent units of selection (Dawkins 1982, pp. 82-89). Apparent group adaptations typically result from gene-level or individual-level selection operating within groups, not selection between groups.

Consider altruism. A gene causing an organism to sacrifice itself for relatives can spread if the beneficiaries share the gene. W.D. Hamilton formalized this as inclusive fitness theory: a gene spreads if it increases the replication of copies of itself, whether in the organism's own offspring or in relatives' offspring (Hamilton 1964). This explains altruism toward kin without invoking group selection. The altruistic gene benefits itself by helping copies in relatives. No group-level explanation is necessary.

The methodological moral: locate adaptation at the level where selection operates most powerfully and where replicators exist. Genes are replicators. Organisms and groups are not. Apparent purposes at higher levels are byproducts of gene competition at lower levels. Teleology is banished from biology not by denying functional organization but by explaining it through differential replication of genes. The eye serves the genes that build it, not the organism that sees with it.

III. The Austrian Solution

The Problem: Institutional Order Without a Designer

Pre-Mengerian economics faced an explanatory crisis parallel to pre-Darwinian biology. Social institutions exhibit intricate functional organization. Money facilitates exchange. Prices coordinate production and consumption. Legal systems resolve disputes. Languages enable communication. The temptation to explain institutional order through state design or enlightened planning was overwhelming.

Mercantilism embodied this constructivist view. Institutions served state purposes. The sovereign designed monetary systems, regulated trade, and imposed laws. Institutional order reflected royal will or legislative intent. Without conscious design from above, chaos would prevail. This doctrine dominated European thought for centuries and persists in contemporary social engineering (Hayek 1988, pp. 35-43).

The Physiocrats and Adam Smith challenged this view but did not fully escape It. Smith recognized that market order emerges unintentionally from individual self-interest: the baker provides bread not from benevolence but from self-interest, yet the outcome serves the common good (Smith 1776). This insight was profound but incomplete. Smith invoked an "invisible hand" guiding individual actions toward collective benefit, a metaphor suggesting providential design. He did not explain the mechanism generating institutional order from individual action. The Austrian economists completed this project.

Menger's Breakthrough: The Spontaneous Origin of Institutions

Carl Menger solved the problem of institutional origins by demonstrating that complex institutions can emerge without anyone designing them (Menger 1871). His paradigm case was money. Every society uses a medium of exchange. Money is not a government invention. No assembly of wise men convened to declare that gold or silver or cowrie shells would serve as money. Money emerged spontaneously.

Menger's account proceeds as follows. Barter requires a double coincidence of wants. A farmer with wheat seeking fish must find a fisherman with fish seeking wheat. This limits exchange. Individuals discover that certain goods are more widely accepted than others. Cattle, salt, or precious metals find ready buyers. A farmer with wheat can trade for cattle even if he does not want cattle, because he can later trade cattle for fish. The cattle serve as an intermediate medium of exchange.

As this pattern spreads, a few goods become universally accepted media. Precious metals dominate because they are durable, divisible, portable, and scarce. Money emerges from countless individuals independently choosing to hold and trade the most marketable goods. No central planner designs the monetary system. *No social contract establishes money. Individuals acting in their own interest unintentionally generate a monetary institution* (Menger 1871, pp. 257-262).

This account generalizes beyond money. Language emerges from individuals imitating successful communicative gestures. Legal norms emerge from individuals resolving disputes through precedent and custom. Markets emerge from individuals seeking mutually beneficial trades. Institutional order arises through a process Menger called "organic": unintended consequences of intentional individual actions (Menger 1883, pp. 146-147).

Mises's Human Action: Purposeful Action, Unintended Order

Ludwig von Mises developed Menger's insight into a general theory of human action and social order (Mises 1949). Human action is purposeful: individuals choose means to achieve ends. Economics studies these purposeful actions and their consequences. But the consequences extend beyond the actor's intentions. Market prices, for instance, result from individual buy and sell decisions, but no individual intends the overall price structure. Social order is an unintended consequence of intentional actions.

Mises emphasized methodological individualism: only individuals act. Groups, states, and institutions do not act. They are conceptual constructs describing patterns of individual action. To explain institutional phenomena, one must trace them to individual choices. Aggregate outcomes emerge from individual decisions interacting through markets, norms, and legal rules. This individualism parallels Williams's and Dawkins's gene-centered individualism: just as adaptations must be explained through gene-level selection, institutional order must be explained through individual-level action.

Mises distinguished between praxeology, or the logic of human action, and history, or the contingent facts about which actions individuals choose (Mises 1949, pp. 1-10). Praxeology yields necessary truths: if individuals prefer more to less and act to achieve their ends, certain outcomes follow. Institutional order is one such outcome. Individuals pursuing private ends generate public goods unintentionally. This parallelism between individual purposes and unintended institutional order is the central puzzle Mises solved.

Hayek's Extended Order: Kosmos Not Taxis

F.A. Hayek completed the Austrian solution by distinguishing two kinds of order and explaining how one emerges without design (Hayek 1973, 1988). Taxis is made order: consciously designed, hierarchically structured, serving specified purposes. An army, a corporation, or a planned economy exhibits taxis. The order reflects the designer's intentions. Kosmos is grown order: spontaneously emergent, polycentric, serving no single purpose. A market, a language, or a common law system exhibits kosmos. The order arises from individuals following abstract rules without anyone designing the overall pattern.

Hayek's central thesis is that complex societies depend on kosmos, not taxis. Market economies coordinate millions of individuals producing and consuming goods without central direction. Prices aggregate dispersed knowledge about supply and demand. Legal systems resolve disputes through evolving precedents and norms. Languages enable communication through grammatical rules no one consciously designed. These institutions exhibit "extended order": coordination extending far beyond any individual's knowledge or control.

Hayek writes: "*The extended order resulted not from human design or intention but from spontaneously conforming to certain traditional and largely moral practices, many of which men tend to dislike, whose significance they usually fail to understand, whose validity they cannot prove, and which have nonetheless fairly rapidly spread by means of an evolutionary selection*" (Hayek 1988, p. 6). This passage encapsulates the Austrian solution. Order emerges from individuals following rules. The rules spread through selection. Individuals need not understand the overall order. Design is absent.

Hayek distinguished between two errors. Constructivist rationalism attempts to design social order from above, imposing taxis on society. This fails because central planners

lack the dispersed knowledge required to coordinate complex economies. Markets process information through prices far more efficiently than any planning agency could (Hayek 1945). The opposite error is refusing to recognize that some institutions can and should be designed. Legal frameworks, constitutions, and property rights require conscious design. *The distinction is between designing rules (permissible) and designing outcomes (impossible)*. Individuals following well-designed rules generate spontaneous order. Attempting to design the order itself produces chaos (Hayek 1973, pp. 45-46).

The Parallel to Dawkins

The terminological parallel between Hayek's "extended order" and Dawkins's "extended phenotype" is striking. Both concepts address the same problem: explaining functional order at a level higher than the acting unit without invoking purposes at that higher level. Dawkins explains how genes produce phenotypic effects beyond the organism's body. Hayek explains how individuals produce institutional effects beyond their intentions. Both reject teleology at the higher level. Both locate the generative mechanism at the lower level.

Hayek did not cite Dawkins, and Dawkins did not cite Hayek. The convergence is independent. This independence strengthens the case for structural identity. Two traditions, addressing different domains, converged on the same solution because the underlying problem is the same. Explaining order without a designer requires locating the generative mechanism in lower-level units following simple rules. The cumulative result is complex functional order that appears designed but is not.

IV. Structural Convergence

Formal Equivalence of the Problems

The pre-Darwinian biological problem and the pre-Mengerian economic problem are formally equivalent. Both involve explaining complex functional order without invoking a designer. In biology, the explanandum is organismal adaptation: eyes that see, hearts that pump, kidneys that filter. In economics, the explanandum is institutional coordination: markets that allocate resources, legal systems that resolve disputes, languages that enable communication. Both domains exhibit intricate functional organization crying out for explanation.

The pre-Darwinian solution invoked divine design or teleological principles: organisms exist for purposes, species serve functions, nature tends toward perfection. The pre-Mengerian solution invoked state design or enlightened planning: institutions serve state purposes, sovereigns design social order, laws reflect legislative will. Both solutions appealed to intentional design at a higher level than the phenomena being explained. God or Nature designed organisms. Kings or Legislators designed institutions.

Darwin and Menger rejected these higher-level designers. Both explained functional order through lower-level mechanisms. Darwin explained adaptation through differential survival and reproduction of individuals. Menger explained institutional order through unintended consequences of individual actions. Both located the generative process at the level of individuals pursuing their own survival or interests. Both showed how complex order emerges from simple rules operating across many iterations. Both banished teleology from their respective domains.

Structural Identity of the Solutions

The Darwinian and Austrian solutions share the same logical structure. Table 1 displays the correspondence.

Table 1: Structural Correspondence Between Dawkins and Hayek

Element	Dawkins	Hayek
Unit of analysis	Gene (replicator)	Individual actor
Generative mechanism	Differential replication	Intentional action
Cumulative process	Natural selection over generations	Market competition and legal evolution
Outcome	Adaptation (apparent design)	Institution (spontaneous order)
Extended effect	Extended phenotype (beaver dam)	Extended order (market prices)
Error rejected	Species/group teleology	Constructivist rationalism
Correct question	Cui bono? (which gene benefits?)	Who acts? (methodological individualism)
Apparent "purpose"	Byproduct of gene competition	Byproduct of individual actions
Higher-level order	Organism, ecosystem	Civilization, market
Why higher-level order is not explanatory	Organisms and groups do not replicate	Institutions do not act

The correspondence is precise. Genes are units of selection. Individuals are units of action. Genes replicate differentially based on their effects. Individuals act purposefully based on their preferences. Cumulative differential replication produces adaptations. Cumulative individual actions produce institutions. Both adaptations and institutions appear designed but are not. Both arise from lower-level mechanisms operating without higher-level purposes.

The concept of extended effects is central to both theories. Dawkins's extended phenotype describes phenotypic effects beyond the organism's body. Hayek's extended order describes institutional coordination beyond any individual's knowledge or intention. Both concepts sever the connection between the acting unit and the boundaries of its effects. Genes affect the world beyond the organism. Individuals affect social order beyond their intentions. Selection operates on extended phenotypes.

Competition and imitation operate on institutions as extended consequences of individual actions.

Both theories reject teleology at higher levels. Dawkins rejects species-level and group-level purposes. Adaptations do not serve species survival or group welfare. They serve gene replication. Asking what benefit a species derives from a trait is the wrong question. The correct question is which genes benefit. Hayek rejects society-level and state-level purposes. Institutions do not serve collective welfare or state power (except incidentally). They emerge from individual actions. Asking what social function an institution serves is the wrong question. The correct question is which individuals' actions sustain the institution and why.

Cui Bono and Methodological Individualism

Dennett's cui bono criterion in biology corresponds precisely to methodological individualism in Austrian economics. Dennett insists that every purported adaptation must be traced to the replication of specific genes. Benefits to organisms or groups do not explain traits unless those benefits enhance gene replication. The question "Who benefits?" must receive a gene-level answer.

Methodological individualism insists that every institutional phenomenon must be traced to the actions of specific individuals. Benefits to society or the state do not explain institutions unless those benefits emerge from individuals pursuing their own ends. The question "Who acts?" must receive an individual-level answer. Groups, classes, and states are conceptual abstractions. Only individuals act (Mises 1949, p. 42).

The two principles share the same logical form. Both demand reduction of higher-level phenomena to lower-level mechanisms. Both reject emergent purposes at higher levels. Both insist that functional organization be explained through competition and selection at lower levels. Dawkins reduces adaptation to gene competition. Hayek reduces institutional order to individual action. Both are methodological reductionists in the same sense.

This reductionism does not deny higher-level order. Organisms exist. Ecosystems exist. Markets exist. Civilizations exist. The claim is that these higher-level entities do not act or have purposes. They are patterns resulting from lower-level processes. Explaining the patterns requires explaining the processes. Adaptations result from gene-level selection. Institutions result from individual-level action. The higher-level order is real but epiphenomenal with respect to explanation.

The Error Structure

Both theories define themselves by rejecting the same error: invoking purposes at levels where purposes do not exist. Pre-Darwinian biology invoked species-level purposes: traits exist for species survival. Wynne-Edwards revived this error with group selection: traits exist for group welfare (Wynne-Edwards 1962). Williams and Dawkins demolished these claims by showing that gene-level selection explains

apparent group adaptations more parsimoniously (Williams 1966; Dawkins 1976, 1982).

Pre-Mengerian economics invoked state-level purposes: institutions serve state power or collective welfare. Contemporary constructivism revives this error: institutions should be designed to serve social justice, equality, or other collective ends. Menger, Mises, and Hayek demolished these claims by showing that institutional order emerges from individual action and cannot be designed from above (Menger 1871; Mises 1949; Hayek 1945, 1973, 1988).

The error structure is identical. Group selection reintroduces species-level or group-level teleology into biology. Constructivism reintroduces society-level or state-level teleology into economics. Both errors assume that higher-level entities have purposes that explain lower-level phenomena. Both errors invert the correct explanatory direction. Genes do not exist to serve species. Individuals do not exist to serve states. The explanatory arrow points from genes to adaptations and from individuals to institutions, not the reverse.

Dawkins and Hayek both emphasize that the error persists because higher-level order appears designed. Organisms look as if they were designed for survival. Institutions look as if they were designed for social welfare. The appearance is misleading. The order results from cumulative lower-level selection, not higher-level design. Recognizing this requires resisting psychological intuitions that attribute purposes where none exist.

V. The Synthesis: Extended Phenotype Theory Applied to Institutions

Hayek's Missing Mechanism

Hayek explained spontaneous order conceptually but not mechanistically. He showed that institutions emerge from individuals following rules and that institutional evolution occurs through selection of successful rules. But he did not specify what is being selected or how selection operates. Rules are abstractions. Individuals follow rules, but rules do not replicate. Hayek's account lacks a unit of selection analogous to the gene.

Dawkins provides the missing element: the meme. Richard Dawkins coined the term "meme" to denote cultural replicators analogous to genes (Dawkins 1976, pp. 189-201). Memes are units of cultural information: ideas, norms, skills, practices. Memes replicate through imitation and communication. A catchy tune is a meme. A religious belief is a meme. A legal precedent is a meme. Memes compete for replication in human minds just as genes compete for replication in gene pools.

Hayek's rules are memes. Legal norms, property rights, linguistic conventions, and market practices are all memes. Individuals adopt and follow these memes. Successful

memes spread. Memes that facilitate coordination, reduce transaction costs, or resolve disputes more effectively outcompete alternatives. Institutional evolution is memetic evolution: differential replication of memes based on their effects on human behavior and social outcomes.

The synthesis completes Hayek's account. Institutions are extended phenotypes of memes. Just as genes build extended phenotypes like beaver dams and spider webs, memes build extended phenotypes like legal systems and market institutions. A legal rule is a meme residing in the minds of judges, lawyers, and citizens. The legal system comprising courts, statutes, and enforcement mechanisms is the extended phenotype of legal memes. The system persists because the memes replicate through legal education, judicial decisions, and normative compliance.

Institutions as Extended Phenotypes of Memes

Consider property rights as an example. The concept of private property is a meme: an idea about how individuals may legitimately acquire, use, and transfer resources. This meme resides in the minds of individuals. Its extended phenotype is the institutional structure protecting property: courts adjudicating disputes, police enforcing judgments, contracts transferring ownership. The entire property system is an extended phenotype of the property-rights meme.

Why does the property-rights meme replicate? Because individuals who adopt it benefit from reduced conflict, increased investment, and expanded trade. Societies where property rights are widely respected experience economic growth. Individuals observe this correlation and adopt the meme. The meme spreads through imitation, education, and institutional reinforcement. The extended phenotype (the property system) facilitates the meme's replication by creating environments where property rights are salient and beneficial.

This analysis applies to all legal institutions. Contract law is an extended phenotype of memes defining enforceable agreements. Tort law is an extended phenotype of memes allocating liability for harms. Constitutional law is an extended phenotype of memes distributing political authority. Each legal domain consists of memes (ideas, norms, precedents) and their extended phenotypes (courts, statutes, enforcement mechanisms).

The extended phenotype framework explains institutional evolution. Memes mutate through innovation: a judge creates a novel precedent, a legislator proposes a new rule, a legal scholar articulates a new principle. Memes are selected based on their effects: rules that reduce transaction costs, facilitate cooperation, or resolve disputes more effectively spread through imitation and adoption. Institutional change is memetic evolution: variation, selection, and replication of legal memes.

Dennett's Cui Bono Applied to Institutions

Applying Dennett's cui bono criterion disciplines institutional analysis. Every institutional phenomenon must be traced to the replication of specific memes. Asking what benefit society derives from an institution is the wrong question. The correct question is which memes benefit from the institution's existence and how those memes replicate.

Consider the common law tradition. The institution generates precedent-based legal rules through judicial decisions. One might ask: what social function does this serve? Stability, predictability, adaptability? These are byproducts. The explanatory question is: which memes benefit from common law institutions? Legal memes that generate useful precedents replicate through citation, adoption by other courts, and legal education. Judges who create influential precedents gain prestige, incentivizing meme innovation. Law schools teach canonical cases, spreading memes across generations. The institution persists because legal memes replicate through it.

The analysis inverts conventional functionalism. Institutions do not exist to serve social functions. Social functions are byproducts of memetic replication. Legal systems stabilize societies, but that is not why they exist. They exist because legal memes replicate through them. The stability is a side effect that incidentally enhances the system's longevity, thereby facilitating further memetic replication. But the primary explanation is memetic, not functional.

This reorientation has radical implications. It undermines claims that institutions should be designed to serve collective welfare or social justice. Such claims assume that institutions exist to serve society-level purposes. But institutions do not have purposes. They are extended phenotypes of replicating memes. Attempting to design institutions to serve abstract collective ends is like attempting to design organisms to serve species survival. The attempt misunderstands the generative mechanism.

Against Multilevel Selection in Biology and Society

The synthesis undermines multilevel selection theory (MLS2) in both biology and social science. MLS2 proponents argue that selection operates simultaneously at multiple levels: genes, individuals, groups (Sober and Wilson 1998; Wilson and Wilson 2007). Groups compete, and traits benefiting groups can evolve even if they harm individuals. This resurrects group-level teleology under a new label.

Dawkins's response applies equally to biological and cultural evolution. Groups do not replicate with sufficient fidelity to function as units of selection. Group boundaries are porous. Individuals migrate between groups. Groups fission and fuse. Cultural groups are even less coherent than biological groups. Ethnic groups, nations, and civilizations constantly exchange members, ideas, and institutions. Treating groups as units of selection in cultural evolution is even less plausible than in biological evolution (Dawkins 1982, pp. 82-89).

The correct unit of cultural selection is the meme. Memes replicate through imitation and communication. Memes compete within minds and across social networks. Successful memes spread. Apparent group-level adaptations result from meme-level selection, not group-level selection. A society that adopts property rights memes prospers, but this does not mean the society as a group is the unit of selection. The property rights memes spread because individuals who adopt them benefit, leading others to imitate.

The error of MLS2 is attributing explanatory power to higher-level selection when lower-level selection suffices. In biology, gene-level selection explains apparent group adaptations. In cultural evolution, meme-level selection explains apparent society-level adaptations. Invoking group selection or society-level selection adds nothing explanatory and reintroduces teleology.

The failure of MLS2 is not merely empirical but conceptual. The theory never specifies the causal mechanism by which one group exerts selective pressure on another. Consider the purported case: a group of altruists outcompetes a group of selfish individuals. What is the causal chain? Groups do not reproduce like organisms. They do not undergo mitosis. A "successful" group does not bud off daughter groups carrying its "genetic" material. Groups grow through recruitment, fragment through fission, and merge through alliance. None of these processes constitute replication in the biological sense. Without replication, there is no inheritance. Without inheritance, there is no cumulative selection.

In contrast, the gene-level mechanism is specifiable. A gene causing altruistic behavior toward kin spreads if recipients share the gene. The causal chain is clear: *gene → behavior → benefit to kin → replication of gene copies in kin → increased frequency of gene in population*. Hamilton's rule formalizes this: a gene for altruism spreads if $rB > C$, where r is relatedness, B is benefit to recipient, and C is cost to actor (Hamilton 1964). No group-level explanation is needed. The apparent group benefit (a population of altruists) is a byproduct of gene-level selection operating through kinship.

The replicator problem is fatal to MLS2. Dawkins defined replicators as entities that make copies of themselves with sufficient fidelity for cumulative selection to operate (Dawkins 1982, pp. 82-89). Genes are replicators: DNA sequences copy themselves through molecular machinery with error rates low enough (approximately 10^{-9} per base pair per generation) for cumulative adaptation. Organisms are not replicators: they do not copy themselves but produce offspring through sexual recombination, which scrambles genetic information. Groups are even less plausible as replicators. A "group" has no mechanism for copying its structure. When a population splits, the two subpopulations do not carry identical copies of group-level information. Cultural groups are worse: they constantly exchange members, adopt foreign practices, and merge with competitors. Treating such fluid entities as units of selection is conceptually incoherent.

The debate over group selection reached its zenith in 2012 when Steven Pinker published "The False Allure of Group Selection" on Edge.org. Pinker argued that group selection is both empirically unsupported and theoretically unnecessary. Gene-level

selection, supplemented by kin selection and reciprocal altruism, explains purported cases of group adaptation without invoking group-level processes. The essay triggered a 41,000-word debate involving the field's leading figures. Richard Dawkins, Daniel Dennett, and John Tooby supported Pinker's gene-centered view. David Sloan Wilson, Peter Richerson, and Robert Boyd defended multilevel selection. The debate clarified that the gene-centered framework has emerged as the consensus position in evolutionary biology. MLS2 proponents have failed to identify a single case where group-level selection provides explanatory power that gene-level selection lacks (Pinker 2012).

The application to cultural evolution is direct. Memes, like genes, are replicators. They copy themselves through imitation and communication. Legal memes spread through citation, education, and precedent. Economic memes spread through imitation of successful practices. Linguistic memes spread through communication and learning. All of these processes occur at the level of individual minds, not groups. A "society" does not replicate its institutional structure when it grows or fragments. Institutions spread because individuals adopt memes and build extended phenotypes. Group selection in cultural evolution suffers from the same conceptual problems as group selection in biological evolution: no replicator, no specified mechanism, no explanatory advantage over individual-level and meme-level selection.

The Epistemological Inaccessibility of the Common Good

The synthesis also undermines appeals to the common good as an objective end. If institutions are extended phenotypes of memes, then institutional functions are byproducts of memetic competition, not designed purposes. There is no common good independent of individual preferences and memetic dynamics. Claims that institutions should serve the common good assume a perspective external to the evolutionary process. No such perspective exists.

Hayek emphasized the epistemological problem: knowledge is dispersed across individuals. No central planner possesses the information required to design institutions serving collective welfare (Hayek 1945). The synthesis radicalizes this point. Even if a planner possessed perfect information, there is no common good to serve. The common good is not an objective feature of reality but an abstraction from individual preferences. Institutions emerge from memetic competition, not from pursuit of collective ends.

This does not imply nihilism or relativism. Individuals have preferences. Memes that satisfy those preferences spread. Institutions that facilitate coordination, reduce conflict, and enable prosperity replicate because individuals adopt the memes sustaining them. But there is no common good over and above this process. The appearance of collective purposes is a byproduct of individual actions and memetic replication.

VI. Implications and Conclusions

Implications for Institutional Design

The Dawkins-Hayek synthesis reorients institutional design. Designers cannot design outcomes. They can only design selective environments where beneficial memes are more likely to replicate. This parallels the distinction in evolutionary biology between designing organisms (impossible) and designing environments where beneficial traits are selected (possible through artificial selection).

Constitutional designers, for instance, cannot design a just society. They can design a constitutional framework that selects for memes promoting stability, liberty, and prosperity. Separation of powers, federalism, and individual rights are institutional structures that favor memes of limited government and individual autonomy. These memes replicate because they generate outcomes individuals prefer. The constitutional framework channels memetic evolution toward beneficial outcomes without dictating those outcomes directly.

This approach differs from constructivism. Constructivists attempt to design institutions that produce specified outcomes: equality, justice, collective welfare. The attempt fails because it misunderstands the generative mechanism. Institutions are extended phenotypes, not designed machines. Attempting to design them to serve collective ends is like attempting to design organisms to serve species survival. The correct approach is designing selective environments where beneficial memes outcompete harmful ones.

Judicial precedent provides an example. Common law systems do not design legal rules directly. They design a selective environment (precedent-based adjudication) where useful legal memes spread through judicial decisions. Judges resolve disputes by applying and refining prior decisions. Useful precedents are cited and followed. Unhelpful precedents are distinguished or overruled. The cumulative result is a body of law adapted to social needs without anyone designing it.

Against Group Selection and Constructivism

The synthesis reinforces the case against group selection in biology and constructivism in social science. Both doctrines reintroduce teleology at levels where teleology does not exist. Group selection attributes purposes to groups. Constructivism attributes purposes to societies or states. Both errors invert the correct explanatory direction.

Biological adaptations serve gene replication, not group survival. Cultural institutions emerge from memetic replication, not collective design. Explaining biological adaptations requires tracing them to gene-level selection. Explaining cultural institutions requires tracing them to individual actions and memetic dynamics. Appeals to group welfare or collective purposes are explanatorily idle.

This does not deny that groups and societies exhibit order. The claim is that the order is epiphenomenal with respect to explanation. Genes generate organisms, which form populations, which constitute ecosystems. Individuals perform actions, which generate institutions, which constitute societies. The higher-level order is real but results from lower-level processes. Explaining the order requires explaining the processes.

Implications for Constitutional Political Economy

Constitutional political economy studies the design and evolution of constitutional rules (Buchanan and Tullock 1962; Brennan and Buchanan 1985). The Dawkins-Hayek synthesis provides a theoretical foundation for this project. Constitutional rules are memes. Constitutional systems are extended phenotypes of those memes. Constitutional design is the engineering of selective environments where beneficial constitutional memes replicate.

Successful constitutional systems exhibit memetic diversity and competition. Federalism allows memes to compete across jurisdictions. Separation of powers allows memes to compete across branches. Judicial review allows legal memes to compete in courts. Amendment procedures allow constitutional memes to evolve. These institutional features facilitate memetic evolution without predetermining outcomes.

Failed constitutional systems suppress memetic competition. Authoritarian regimes impose a single set of memes and prevent alternatives from replicating. Centralized planning attempts to design outcomes rather than allowing institutions to emerge. Both approaches fail because they prevent the trial-and-error process through which beneficial memes are discovered. The lesson is that constitutional design should maximize memetic diversity and competition, not predetermine substantive outcomes.

Open Questions and Ongoing Research

The synthesis developed here provides the theoretical foundation for Extended Phenotype Theory (EPT), a research program I have been developing across multiple empirical applications. Several questions merit continued investigation.

First, how precisely does memetic evolution differ from genetic evolution? Memes mutate faster, spread horizontally, and blend in ways genes do not. Preliminary work suggests these differences generate faster institutional evolution and greater cross-jurisdictional diversity than biological evolution exhibits. Characterizing these dynamics formally remains ongoing.

Second, what are the units of memetic selection? Empirical analysis of constitutional clauses suggests a hybrid answer: individual provisions function as atomic memes, but they cluster into memeplexes (doctrinal complexes) that replicate together. The Constitutional Lock-in Index operationalizes this insight by measuring both clause-level entrenchment and network-level path dependence (Lerer 2025b).

Third, how do memes interact? Genes interact epistatically; memes likely do as well. The IusMorfos metric addresses this by measuring morphological compatibility between institutional configurations, predicting transplant success based on memetic complementarity rather than isolated trait analysis.

Fourth, empirical testing has progressed substantially. The Constitutional Lock-in Index achieves 78% predictive accuracy across 47 reform attempts. Genealogical analysis of 120 constitutional clauses confirms gradual normative evolution (91%) with punctuational architectural innovation (19%). These results, detailed in companion papers, validate EPT's core predictions against MLS2 alternatives.

Conclusion: Two Traditions, One Solution

Charles Darwin, Richard Dawkins, Carl Menger, Ludwig von Mises, and F.A. Hayek independently solved the same problem: explaining complex functional order without invoking a designer. The biological tradition explained adaptation through gene-level selection and extended phenotypes. The economic tradition explained institutional order through individual action and spontaneous order. The convergence is not analogical but structural. The two solutions share the same logical architecture because they address formally equivalent problems.

The synthesis completes both traditions. Dawkins's concept of the extended phenotype provides the mechanistic foundation Hayek's extended order lacked. Institutions are extended phenotypes of memes competing for replication. Hayek's concept of spontaneous order provides the social application Dawkins's memetics needed. Legal systems, markets, and languages are extended phenotypes of legal, economic, and linguistic memes.

The synthesis has profound implications. It undermines group selection in biology and constructivism in social science as resurrections of pre-Darwinian and pre-Mengerian errors. It reorients institutional design toward engineering selective environments rather than designing outcomes. It provides a theoretical foundation for constitutional political economy and legal evolution.

I have argued that the Dawkins-Hayek convergence reveals a deep theoretical unity spanning biology and social science. Two traditions, working independently, converged on the same solution because the structure of the problem demanded it. Explaining order without a designer requires locating the generative mechanism at the level of replicating units following simple rules. In biology, the units are genes. In culture, the units are memes. In both domains, complex functional order emerges from cumulative selection without higher-level purposes or design.

The Extended Phenotype Theory developed here extends Dawkins's biological insight to legal systems. Legal institutions are extended phenotypes of legal memes. Constitutional systems are extended phenotypes of constitutional memes. Institutional evolution is memetic evolution: variation, selection, and replication of legal ideas across

time and space. This framework disciplines institutional analysis by rejecting group-level teleology and emphasizing individual action and memetic competition. It establishes the theoretical foundation for a research program analyzing legal evolution through the same conceptual tools evolutionary biology uses to analyze organic evolution.

This framework has been applied to constitutional lock-in, precedent dynamics, legal transplants, and institutional competition in companion papers. Ongoing work continues refining Extended Phenotype Theory through expanded empirical tests and comparative legal analysis.

The Dawkins-Hayek convergence is not the endpoint but the beginning. It opens a path toward a genuinely evolutionary social science grounded in the same methodological principles that revolutionized biology: gene-level selection, extended phenotypes, and the rejection of teleology. The revolution Darwin began in 1859 and Dawkins radicalized in 1982 can now extend to the analysis of legal and political institutions. The problem of order without design admits a single solution. Understanding that solution illuminates both the natural and the social worlds.

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